

Classifying Network Attacks by Category

*Final Assignment, Supervised Machine Learning Classification course, IBM Machine Learning Professional Certificate
Dataset: NSL-KDD (public network intrusion benchmark, official train/test split)*

Main Objective

This project builds a supervised classifier that labels a network connection as normal traffic or as one of four attack categories: DoS, Probe, R2L, or U2R. For a security operations stakeholder, the goal is a model that flags the right category quickly, especially for rare attack types that are easy to miss among a much larger volume of normal traffic.

Three classification algorithms are compared under the same conditions: Logistic Regression, Random Forest, and Gradient Boosting.

Dataset Description

The dataset is NSL-KDD, the same public network intrusion benchmark used in an earlier unsupervised learning project, now used with its official train/test split for supervised classification. The training set has 125,973 labeled connections, the test set has 22,544. Each connection has 41 features covering protocol, service, connection duration, byte counts, and traffic statistics. The 3 categorical features (protocol_type, service, flag) were one-hot encoded, and the 38 numeric features were scaled.

The 22 original attack labels are grouped into the standard 4 categories used in the NSL-KDD literature (DoS, Probe, R2L, U2R), plus normal traffic.

An important detail: the official test set includes 17 attack labels that never appear in training at all, for example apache2, mailbomb, and sqlattack. This makes the test set a genuine generalization check, not just a random split of attacks the model has already seen.

Category	Train records	Test records
normal	67,343	9,711
DoS	45,927	7,460
Probe	11,656	2,421
R2L	995	2,885
U2R	52	67

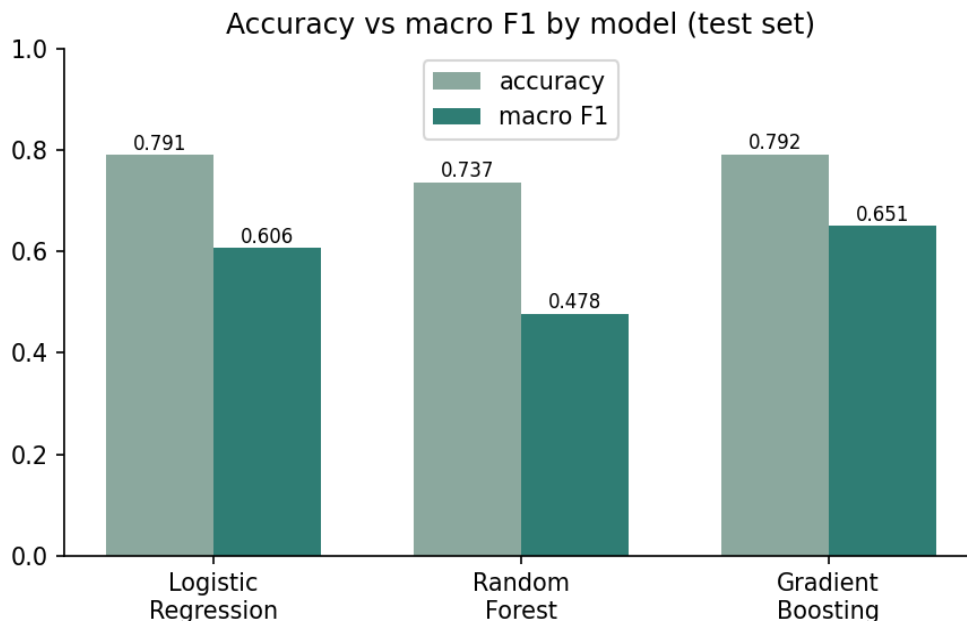
R2L is actually more common in the test set than in training. Combined with the many novel R2L attack labels mentioned above, this category is where the train and test distributions differ the most, which matters later when interpreting the results.

Model Variations

All three models were trained on the same features, with sample weights that upweight rare categories, since normal and DoS traffic massively outnumber R2L and U2R attacks in training (67,343 normal records versus 52 U2R records).

Model	Accuracy	Macro F1
Logistic Regression	79.13%	0.606
Random Forest (200 trees)	73.70%	0.478
Gradient Boosting	79.16%	0.651

Macro F1, not accuracy, is the metric that matters most here. It scores every category equally regardless of how many examples it has, so it does not let strong performance on normal and DoS traffic hide poor performance on rare attacks, the way plain accuracy can.



Final Model Recommendation

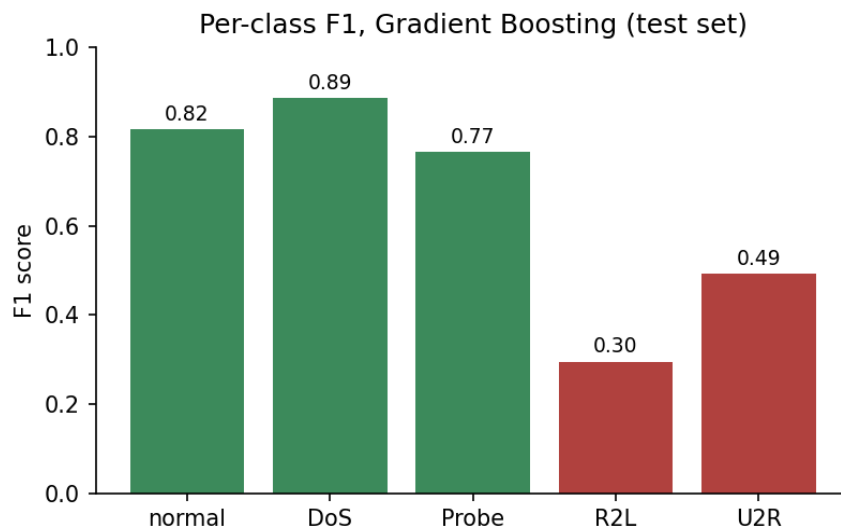
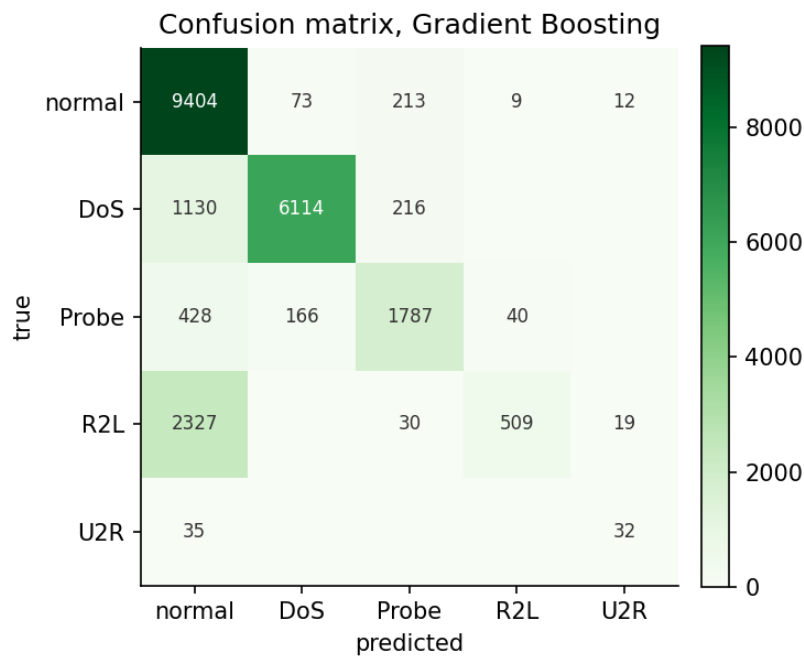
Gradient Boosting is recommended. It has the highest macro F1 (0.651) and ties Logistic Regression on accuracy (79.16% versus 79.13%), which means it reaches the same overall correctness while doing noticeably better across all five categories, not just the common ones.

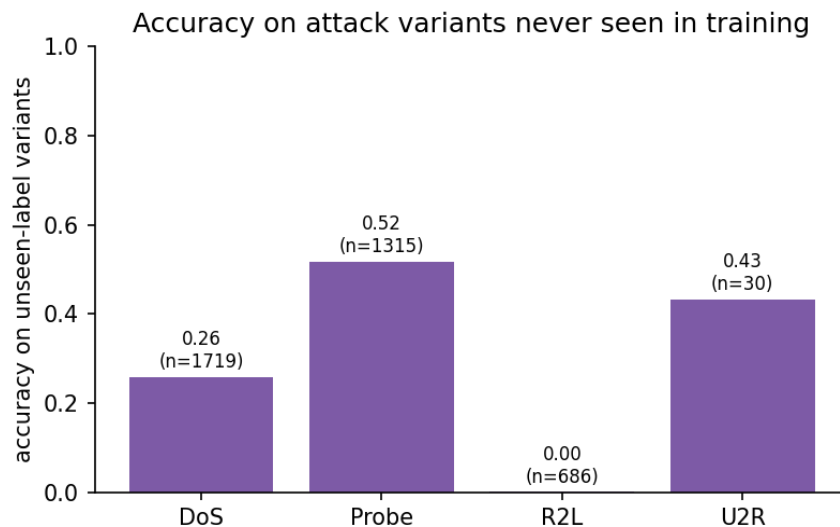
Random Forest is not recommended, despite being a generally strong algorithm. It nearly fails to detect R2L attacks at all (1% recall) even with the same balanced sample weights given to the other two models. This is a specific, real weakness for this dataset, not a generic imbalance problem, since both other models handle R2L far better with the same weighting.

Key Findings and Insights

- Random Forest, despite 200 trees, detects almost no R2L attacks (1% recall) even after reweighting for class imbalance. Logistic Regression and Gradient Boosting both do meaningfully better, so the issue looks specific to how Random Forest handles this particular rare category, not a generic effect of class imbalance.

- Logistic Regression's U2R recall (55%) looks good on paper, but its precision is only 8%. In practice, the model raises a U2R alarm far more often than it is actually right, so it is not a trustworthy detector for that category despite the decent recall number alone.
- Gradient Boosting's U2R result is the most balanced of the three (51% precision, 48% recall). Getting both sides of that trade-off close together at all is a real result, given only 52 U2R examples in the entire training set.
- Testing separately on the attack labels never seen in training shows generalization is not even across categories: 52% accuracy on unseen Probe variants, 43% on unseen U2R variants, only 26% on unseen DoS variants, and effectively 0% on unseen R2L variants. Detecting brand-new R2L techniques is this model's clearest blind spot.





Limitations

- The severe class imbalance (52 U2R examples out of 125,973 training rows) makes it hard for any model to learn a reliable decision boundary for that category, even with balanced sample weights.
- The test set's built-in novel attack labels make this a harder, more honest evaluation than a random split would be, but part of the reported accuracy also reflects attacks the model was never going to recognize by design.
- No hyperparameter tuning was done beyond default settings, aside from adding sample weights. All three models are likely undertuned, and a proper search could change the ranking, especially for Random Forest.
- NSL-KDD is a well-established benchmark, but it is old. A model tuned on it is not guaranteed to transfer to current network traffic.

Next Steps and Future Work

- Tune hyperparameters (tree depth, learning rate, regularization strength) with cross-validation instead of relying on default settings.
- Investigate why Random Forest specifically fails on R2L, for example by comparing its feature importances to what Gradient Boosting picks up on the same data.
- Try a model dedicated to the rare classes, such as a binary R2L-versus-everything classifier or an oversampling technique like SMOTE, instead of asking one model to handle a 1,300-to-1 class ratio directly.
- Feed the confusion patterns back into the SIEM detection rules built in an earlier project, since the categories the model confuses are exactly where a human analyst should double check an automated alert.